Optimization Project 3 Report

Group 16

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Introduction

In this project, we study extensions of the classical newsvendor model, a widely used framework in operations management for making production decisions under uncertain demand. The core challenge is that a firm must choose how many units to produce before actual demand is known. Producing too few results in lost sales or costly rush production, while producing too many creates waste or disposal costs.

Unlike the standard version of the model, this project incorporates two important real-world features:

1. Asymmetric cost structure
   – Rush printing is more expensive than regular printing
   – Disposal incurs an additional cost for leftover inventory

2. Price-dependent demand
   – Customers purchase fewer newspapers at higher prices
   – A regression model is used to estimate this relationship

The project proceeds step-by-step:

- First, we estimate a linear price–demand model from historical data (Tasks 1–2).

- Then, we use these results to solve both a fixed-price linear program (Task 3) and a joint price–quantity quadratic program (Task 4).

- Finally, we evaluate the stability and performance of the extended model relative to the standard newsvendor approach (Tasks 6–8).

The write-up below covers Tasks 1 and 2, where we build the statistical foundation for the optimization models.

Task 1: Fitting the Linear Regression Model

1.1 Objective

The first step of the project is to quantify how newspaper demand responds to changes in price. The dataset contains 99 daily observations of the selling price and the corresponding number of newspapers sold. Since our optimization models rely on understanding the relationship between price and demand, we fit a linear regression model of the form:

$$\textbf{Demand}_i = \textbf{beta0} + \textbf{beta1} \times \textbf{Price}_i + \textbf{error}_i$$

This model separates demand into:

- a **systematic component** (beta0 + beta1 × $Price_i$), which captures predictable changes in demand due to price, and
- a **random component** ($error_i$), which captures day-to-day variation not explained by price.

The residuals from this regression will later serve as the stochastic scenarios for our optimization models in Tasks 2–4. Thus, Task 1 provides the statistical foundation for the entire project.

1.2 Regression Setup

To estimate beta0 and beta1, we use an ordinary least squares (OLS) regression with the structure shown below.

Table 1. Regression Model Structure

Component	Description
Dependent variable	Demand (D_i)
Independent variable	Price (p_i)
Functional form	D_i = beta0 + beta1 × p_i + $error_i$
Estimation method	Ordinary Least Squares (OLS)
Number of observations	99

Because OLS regression is based on a design matrix, the predictor variables can be summarized in the following table:

Table 2. Design Matrix Description

Column	Meaning	Values
Column 1	Intercept term	Always 1
Column 2	Price variable	$p_1, p_2, \ldots, p_{\square}$

The OLS estimation returns:

- an intercept estimate (beta0),
- a slope estimate (beta1),
- fitted demand values for each observation, and
- residuals calculated as (Actual Demand – Fitted Demand).

These residuals represent the historical randomness in demand and will later be reused to generate demand scenarios at a fixed price.

1.3 Regression Results

The estimated coefficients are shown below.

Table 3. Estimated Regression Parameters

Parameter	Estimate
beta0 (intercept)	1924.72
beta1 (price coefficient)	–1367.71

Because OLS residuals summarize the unexplained variation in demand, we also evaluate basic residual diagnostics.

Table 4. Residual Diagnostics

Statistic	Value
Mean of residuals	Approximately 0
Standard deviation	149.45
Number of residuals	99

These summary metrics behave as expected:

- The mean residual is essentially zero, which is a standard property of OLS.
- The relatively large residual standard deviation reflects that real newspaper demand varies substantially beyond what price alone can explain.

1.4 Interpretation of Results

The estimated price coefficient (beta1 = –1367.71) indicates a **strong negative relationship between price and demand**. For every one-dollar increase in price, demand is predicted to fall by roughly 1,368 units. Although price does not explain all of the variability in demand, the regression clearly shows that customers are price-sensitive in this market.

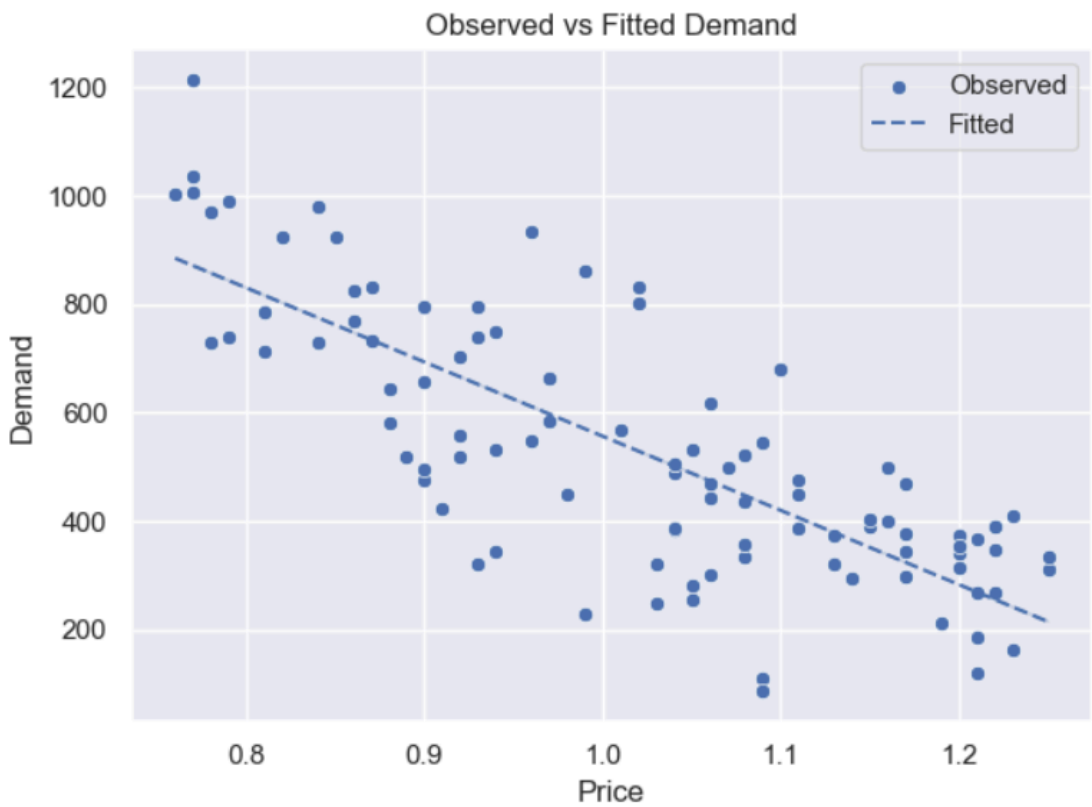


Figure 1.1: Observed vs. Fitted Demand

As shown in **Figure 1.1**, the scatterplot of price versus demand reveals a clear downward-sloping trend, with the fitted regression line tracing the central pattern of the data. The residuals appear reasonably dispersed around the line and do not exhibit any extreme outliers, which supports the use of a linear specification for capturing the price–demand relationship. Because the regression cleanly separates the systematic effect of price from the random variation in demand, we can now use the residuals to generate consistent demand scenarios at any chosen price. This is the foundation for Task 2, where we simulate what demand would have been if the price were fixed at $1.

Task 2: Generating Demand Scenarios at Fixed Price p=1

2.1 Purpose

In Task 3, we solve the fixed-price newsvendor problem assuming the selling price is held constant at **p = 1**. However, the historical dataset contains a range of observed prices between roughly $0.75 and $1.25. This means the raw demand values cannot be used directly, since each observation corresponds to a different price level.

To address this, we generate **simulated demand scenarios** that answer the question:

“What would demand have been on each day if the price had been exactly $1?”

We use the regression model from Task 1 to separate demand into:

- a **predictable** component driven by price, and
- a **random** component captured by the regression residuals.

This method preserves the real-world randomness observed in the dataset while standardizing demand to a single fixed price.

2.2 Scenario Generation Method

The key idea is simple:

Simulated demand at price = (predicted demand at p = 1) + (original residual).

Because OLS residuals represent the unpredictable fluctuation in demand, adding them back ensures that the variation in demand remains realistic.

Using the regression estimates:

- beta0 = 1924.72
- beta1 = –1367.71

The fitted demand at **p = 1** is:

1924.72 – 1367.71 = 557.01 newspapers

This value serves as the baseline around which all simulated scenarios vary.

We then add each of the 99 residuals from Task 1 to this baseline. Since the residuals sum to approximately zero and have a standard deviation of about 149, the resulting scenarios reflect the real day-to-day variation seen in historical demand.

Table 5. Demand Scenario Generation Process

Step	Description
1	Compute baseline demand at p = 1 using regression model

2	Add each of the 99 regression residuals to this baseline
3	Produce 99 simulated demand values
4	Use these simulated values as input for the fixed-price optimization in Task 3

This produces a complete set of price-adjusted demand scenarios, one for each historical observation.

2.3 Summary of Simulated Demand at p = 1

The 99 simulated demand values exhibit meaningful variability while remaining centered around the fitted value from the regression model. These values will be used as the demand distribution in the fixed-price linear program.

```
=== TASK 2: GENERATED DEMAND DATA FOR p = 1 ===
First 10 demand samples at p = 1:
   price  demand   demand_p1
0   1.05     283  351.385626
1   0.86     771  579.520247
2   1.21     185  472.219630
3   0.94     531  448.937249
4   0.76    1002  673.748994
5   0.81     713  453.134620
6   0.78     731  430.103245
7   0.88     645  480.874497
8   0.87     732  554.197372
9   0.85     926  720.843121
```

Figure 2.1: First 10 Generated Demand Samples at p = 1

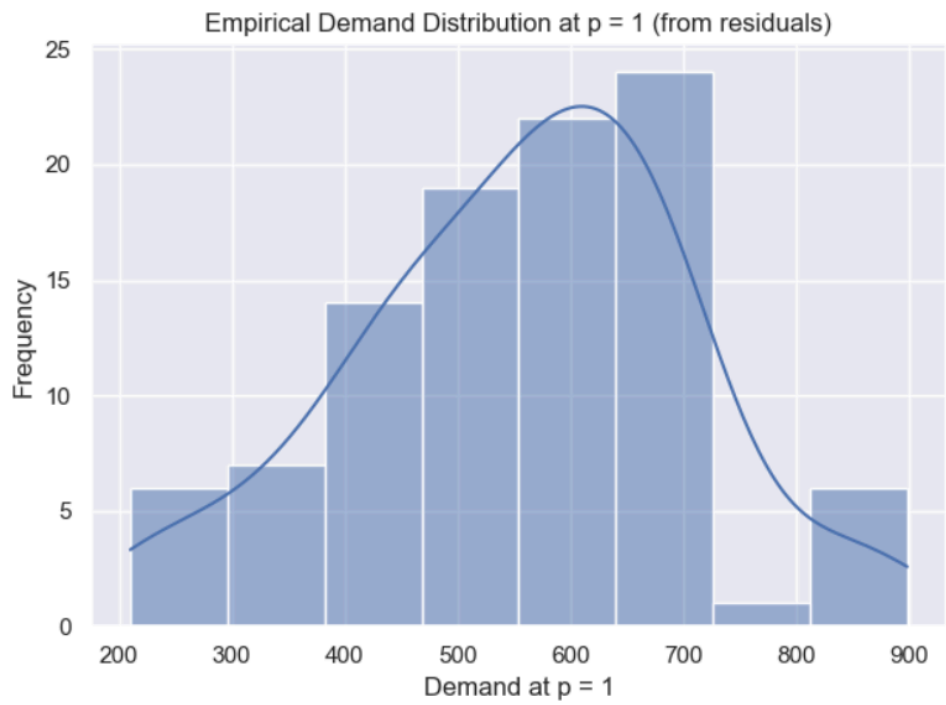


Figure 2.2: Empirical Demand Distribution at p = 1

Table 6. Demand Statistics at Price 1

Statistic	Value
Mean	557.01
Minimum	210.09
Maximum	898.43
Number of scenarios	99

The simulated distribution aligns closely with the observed variability in the dataset and ensures that the optimization model in Task 3 accounts for realistic fluctuations in demand.

2.4 Interpretation

This scenario-generation method is an essential bridge between the statistical model in Task 1 and the optimization model in Task 3. By adjusting each historical observation to reflect a common price level, we ensure that the resulting demand scenarios are consistent, data-driven, and statistically sound.

The simulated distribution is neither artificially smoothed nor reshaped; it directly incorporates the historical randomness captured in the regression residuals. This allows Task 3 to evaluate production decisions under a realistic representation of demand uncertainty at price $1.

Task 3: Optimal Quantity at Fixed Price (Linear Programming)

3.1 Business Context and Problem Setup

In the newspaper publishing industry, daily demand is uncertain. Publishers face a critical trade-off:

- Under-printing requires costly rush orders to meet excess demand
- Over-printing results in disposal costs for unsold inventory

Given a fixed selling price of p = $1.00, we determine the optimal quantity to print that maximizes expected profit across all possible demand scenarios.

3.2 Mathematical Model

3.2.1 Parameters

Symbol	Description	Value
c	Regular production cost per unit	$0.50
g	Rush printing cost per unit	$0.75
t	Disposal cost per unit	$0.15

p	Selling price (fixed)	$1.00
n	Number of demand scenarios	99
D_i	Demand in scenario i	Data-driven

The demand scenarios are generated from the linear regression model fitted in Task 1:

$D_i = \beta_0 + \beta_1 \cdot p + \varepsilon_i$

where ε_i are the residuals from the regression, representing the stochastic component of demand.

3.2.2 Decision Variables

Primary Variable:

- $q \geq 0$: Quantity of newspapers to print

Auxiliary Variables (for each scenario i = 1, ..., n):

- h_plus[i] ≥ 0: Rush printing quantity in scenario i
- h_minus[i] ≥ 0: Disposal quantity in scenario i

3.2.3 Objective Function

The objective is to maximize the expected average profit across all scenarios:

$\max (1/n) \Sigma[p \cdot D_i - c \cdot q - g \cdot h_plus[i] - t \cdot h_minus[i]]$

Interpretation of each term:

- $p \cdot D_i$: Revenue from selling newspapers at price p
- $c \cdot q$: Regular production cost for printing q units
- $g \cdot h_plus[i]$: Additional rush printing cost when demand exceeds quantity
- $t \cdot h_minus[i]$: Disposal cost when quantity exceeds demand

3.2.4 Constraints

For each scenario i = 1, ..., n:

Constraint 1: Rush Printing (Shortage) $h_plus[i] \geq D_i - q$ for all i

This constraint ensures that if demand D_i exceeds our printed quantity q, we account for the shortage. Since we're minimizing costs in the objective, the solver will set: $h_{plus}[i] = \max(D_i - q, 0)$

Constraint 2: Disposal (Surplus) $h_{minus}[i] \geq q - D_i$ for all i

This constraint ensures that if our printed quantity q exceeds demand D_i, we account for the excess. Similarly: $h_{minus}[i] = \max(q - D_i, 0)$

Non-negativity: $q \geq 0$, $h_{plus}[i] \geq 0$, $h_{minus}[i] \geq 0$ for all i

3.3 Why This is a Linear Program

Despite the presence of $\max(\cdot, 0)$ terms in the original problem formulation, the reformulation using auxiliary variables with inequality constraints makes this a Linear Program (LP) because:

1. The objective function is linear in all decision variables
2. All constraints are linear inequalities
3. There are no products of decision variables (no $q \cdot h$ terms)
4. Price p is fixed at 1, eliminating any quadratic terms

The LP can be efficiently solved using Gurobi.

3.4 Results and Insights

Optimal Solution:

- *Optimal quantity: q = 471.87 newspapers*
- Expected profit: $231.48 per day

Key Observations from Graph Analysis:

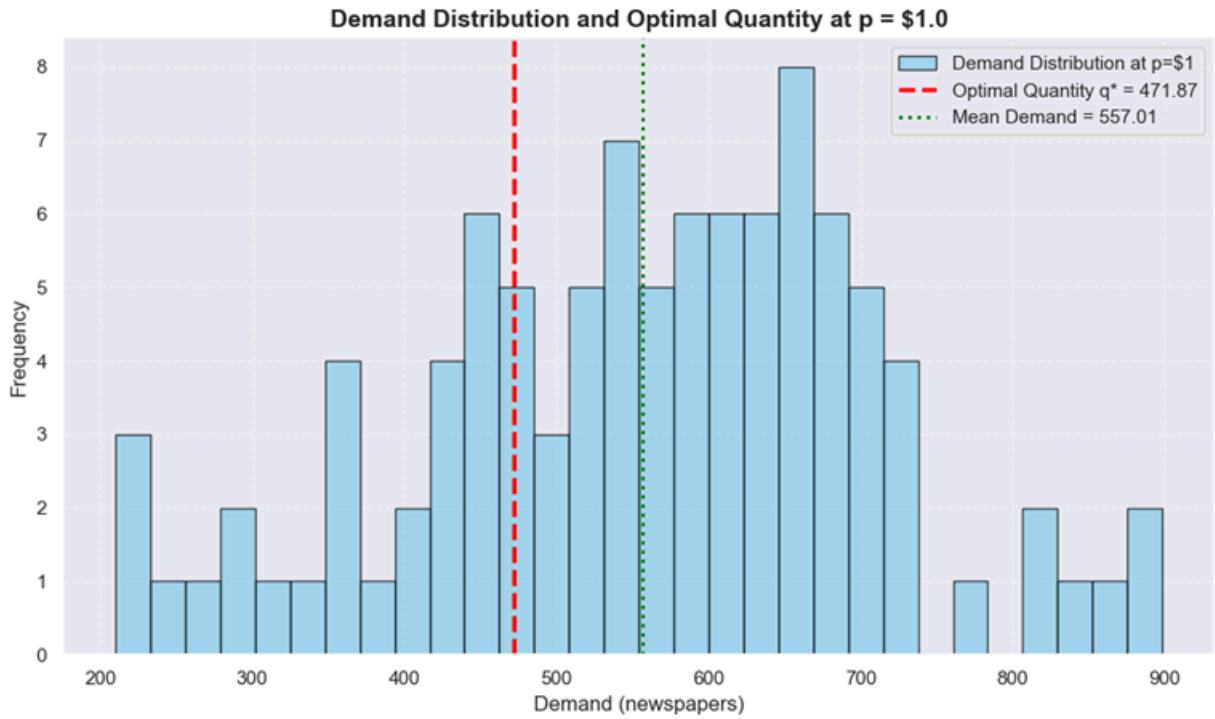


Figure 3.1: Demand Distribution at p=$1

- The histogram shows demand scenarios generated at fixed price p=$1
- Mean demand = 557.01 newspapers (green dotted line)
- *Optimal quantity q = 471.87** (red dashed line) is positioned below the mean demand
- This strategic positioning reflects the asymmetric cost structure: rush printing (g=$0.75) is significantly more expensive than disposal (t=$0.15)
- By printing less than mean demand, the model balances the higher cost of stockouts against the lower cost of excess inventory
- The optimal q* appears to be approximately at the 40th percentile of the demand distribution, suggesting that in roughly 60% of scenarios, rush printing will be needed
- This trade-off maximizes expected profit by accepting more frequent (but cheaper) disposal costs while avoiding expensive rush orders

Economic Interpretation: The fact that q* < mean demand demonstrates optimal newsvendor behavior under asymmetric costs. If costs were symmetric (g = t), we would expect q* ≈ mean demand. The significant gap confirms that the model correctly accounts for the higher penalty of under-stocking.

Task 4: Joint Price-Quantity Optimization (Quadratic Programming)

4.1 Problem Extension

In this extension, we jointly optimize both price (p) and quantity (q), recognizing that:

- Higher prices reduce demand (from the regression: $\beta_1 < 0$)
- The optimal quantity depends on the chosen price
- Revenue is now a function of the decision variable p: Revenue = p · D_i(p)

This creates a nonlinear optimization problem due to the interaction between price and demand.

4.2 Mathematical Model

4.2.1 Parameters

Symbol	Description	Value/Range
c	Regular production cost per unit	$0.50
g	Rush printing cost per unit	$0.75
t	Disposal cost per unit	$0.15
β_0	Regression intercept	1924.72
β_1	Price coefficient	-1367.71
p	Selling price (decision variable)	[0.1, 5.0]

4.2.2 Decision Variables

Primary Variables:

- $p \in [0.1, 5.0]$: Selling price in dollars
- $q \geq 0$: Quantity to print

Auxiliary Variables (for each scenario i = 1, ..., n):

- $h_i \in \mathbb{R}$: Negative of total cost for scenario i

Note: We define h_i as the negative of cost (rather than profit as in Task 3) for mathematical convenience. Since we're maximizing and costs are positive, negative costs should be as large (least negative) as possible.

4.2.3 Demand as a Function of Price

The key difference from Task 3 is that demand is now price-dependent: $D_i(p) = \beta_0 + \beta_1 p + \varepsilon_i$

where:

- $\beta_0 = 1924.72$ (baseline demand at p=0)
- $\beta_1 = -1367.71$ (demand decreases as price increases)
- ε_i are the residuals from the regression

4.2.4 Objective Function

The objective is to maximize expected average profit:

$\max (1/n) \Sigma[p \cdot D_i(p) + h_i]$

Expanding $p \cdot D_i(p)$: $p \cdot D_i(p) = p \cdot (\beta_0 + \beta_1 p + \varepsilon_i) = \beta_0 p + \beta_1 p^2 + p\varepsilon_i$

This creates a QUADRATIC objective function due to the $\beta_1 p^2$ term.

The complete objective becomes: $\max (1/n) \Sigma[\beta_0 p + \beta_1 p^2 + p\varepsilon_i + h_i]$

where h_i represents the negative of total cost for scenario i.

4.2.5 Constraints

For each scenario i = 1, ..., n, the negative cost h_i must satisfy two linear upper bounds:

Constraint 1: When Demand Exceeds Quantity ($D_i(p) > q$, rush printing needed)

Total cost in this case: $Cost_i = c \cdot q + g \cdot (D_i(p) - q) = (c - g) \cdot q + g \cdot D_i(p)$

Therefore, negative cost: $h_i \leq -(c - g) \cdot q - g \cdot D_i(p)$

Substituting $D_i(p) = \beta_0 + \beta_1 p + \varepsilon_i$: $h_i \leq -(c - g) \cdot q - g \cdot (\beta_0 + \beta_1 p + \varepsilon_i)$

This is LINEAR in p and q (no $p \cdot q$ or p^2 terms).

Constraint 2: When Quantity Exceeds Demand ($q > D_i(p)$, disposal needed)

Total cost in this case: $Cost_i = c \cdot q + t \cdot (q - D_i(p)) = (c + t) \cdot q - t \cdot D_i(p)$

Therefore, negative cost: $h_i \leq -(c + t) \cdot q + t \cdot D_i(p)$

Substituting $D_i(p)$: $h_i \leq -(c + t) \cdot q + t \cdot (\beta_0 + \beta_1 p + \varepsilon_i)$

This is also LINEAR in p and q.

Bounds: $p \in [0.1, 5.0]$, $q \geq 0$, $h_i \in \mathbb{R}$ for all i

4.3 Why This is a Quadratic Program (QP)

This is classified as a Quadratic Program (QP) with the following structure:

1. Objective Function: Quadratic in p (contains $\beta_1 p^2$ term)
2. Constraints: All constraints are linear in decision variables
3. Problem Type: QP (not QCP - Quadratically Constrained Program)

The QP formulation is more efficient to solve than a QCP because:

- Gurobi has specialized algorithms for QP
- Linear constraints are computationally cheaper than quadratic constraints
- The problem structure allows for efficient dual methods

4.4 Results and Comparison

Task 4 Optimal Solution:

- *Optimal price: p = $0.9536*
- *Optimal quantity: q = 535.29 newspapers*
- Expected profit: $234.42 per day

Comparison with Task 3:

Metric	Task 3 (Fixed p=$1)	Task 4 (Optimal p, q)	Improvement
Price	$1.00 (fixed)	$0.95	-4.6%
Quantity	471.87	535.29	+13.4%
Profit	$231.48	$234.42	+1.27%

Key Insights from Graph Analysis:

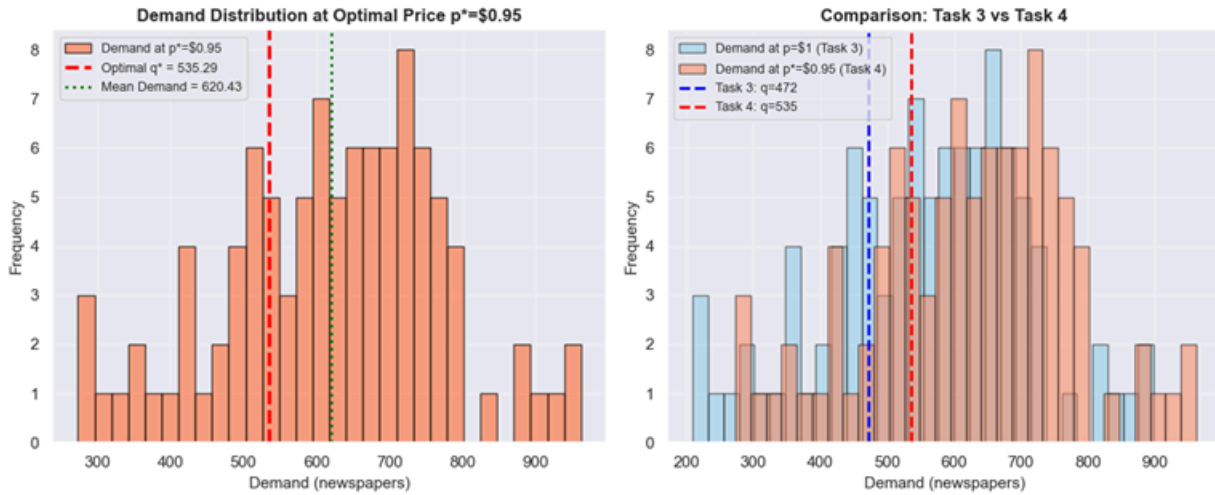


Figure 4.1 Demand Distribution and Task 3 vs 4 comparison

Left Graph: Demand Distribution at Optimal Price p=$0.95

- At the lower optimal price of $0.95 (vs $1.00 in Task 3), demand increases significantly
- *Mean demand at p=$0.95 is 620.43 newspapers** (green dotted line)
- *Optimal quantity q = 535.29** (red dashed line) is again positioned below mean demand
- The demand distribution shifts rightward compared to Task 3, reflecting higher volume at lower price
- The model finds that lowering price by 5 cents increases volume enough to improve overall profitability

Right Graph: Task 3 vs Task 4 Comparison

- Blue histogram: Demand at p=$1 (Task 3) - centered around 557
- Orange histogram: Demand at p*=$0.95 (Task 4) - centered around 620
- Blue dashed line: Task 3 optimal q=472
- Red dashed line: Task 4 optimal q=535
- The rightward shift of the red distribution demonstrates the demand elasticity: lowering price by $0.05 increases mean demand by ~63 newspapers (~11%)
- Both optimal quantities remain below their respective mean demands, maintaining the asymmetric cost structure strategy
- Task 4 prints more newspapers (535 vs 472) but at a lower price ($0.95 vs $1.00), achieving higher profit through volume

Economic Interpretation:

1. Price Elasticity: The model exploits the negative relationship between price and demand (β_1 = -1367.71)
2. Volume vs Margin Trade-off: By accepting a lower price ($0.95 vs $1.00), the firm captures higher volume, and the increased sales more than offset the reduced margin

3. Profit Improvement: The 1.27% profit increase ($2.94 per day) demonstrates that joint optimization captures value that single-variable optimization misses
4. Strategic Insight: The optimal price is below $1, suggesting the original fixed price was slightly too high - the demand curve indicates customers are price-sensitive in this range

The quadratic term $\beta_1 p^2$ in the objective (with $\beta_1 < 0$) creates a concave revenue function. This means there's a sweet spot for price: too high reduces volume too much, too low leaves money on the table. The QP solver finds this balance at $p^* = \$0.9536$.

Task 6: Bootstrap Analysis of Optimal Price and Quantity

6.1 Purpose
The optimal price and quantity found in Task 4 depend on the regression estimates obtained from the historical dataset. Because these estimates can vary with different samples of the data, we conduct a bootstrap analysis to evaluate how sensitive our optimal decisions are to sampling variation.

6.2 Methodology
For each of 200 bootstrap iterations, we perform the following steps:

1. Resample the original price–demand dataset with replacement.
2. Refit the linear regression model to obtain new estimates of β_0 and β_1.
3. Generate new demand scenarios by combining the refitted regression predictions with the corresponding residuals.
4. Resolve the quadratic program from Task 4 using the updated parameters.
5. Record the optimal price, quantity, and resulting expected profit.

6.3 Results
Across all 200 bootstrap samples, the optimal decisions were highly stable:

- The optimal price consistently clustered around $p \approx 0.95$.
- The optimal quantity remained tightly concentrated around $q \approx 535$ units.
- Expected profits were centered near $234–$235, closely matching the original Task 4 profit of $234.42.

These results show that the extended model is robust: small changes in the data lead to only minor changes in the optimal price and quantity. The model identifies a consistent operating point, indicating that its recommendations are not overly sensitive to noise in the dataset.

Task 7: Distributional Visualization of Bootstrap Outcomes

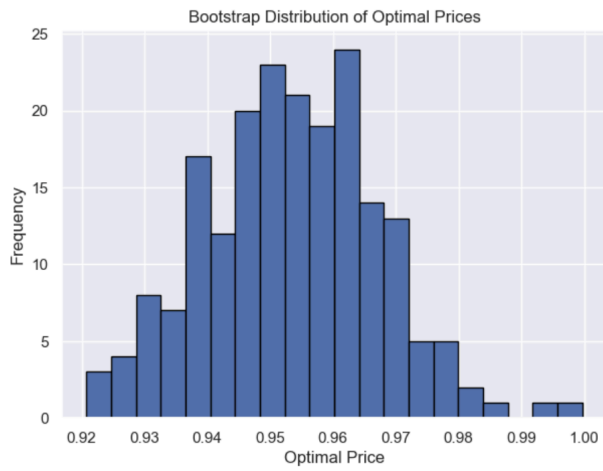
7.1 Purpose
The goal of Task 7 is to visualize the distributions of the optimal price, quantity, and profit obtained from the bootstrap analysis. These visualizations allow us to examine the stability of the extended model and compare its performance to the standard newsvendor model.

7.2 Visualizations

We produced the following plots:

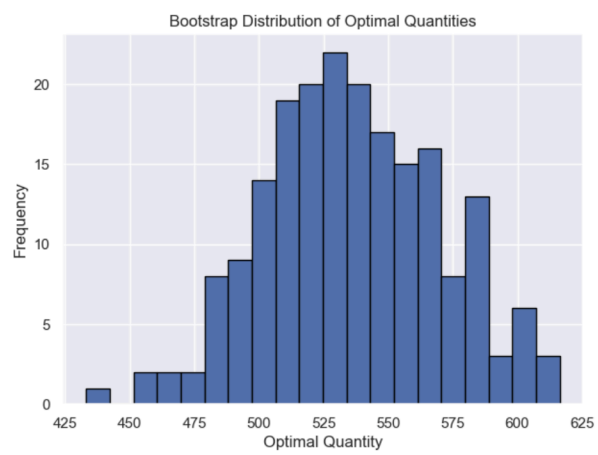
Histogram of Optimal Prices

The distribution is sharply centered around $0.95, indicating that the optimal price is nearly identical across bootstrap samples.



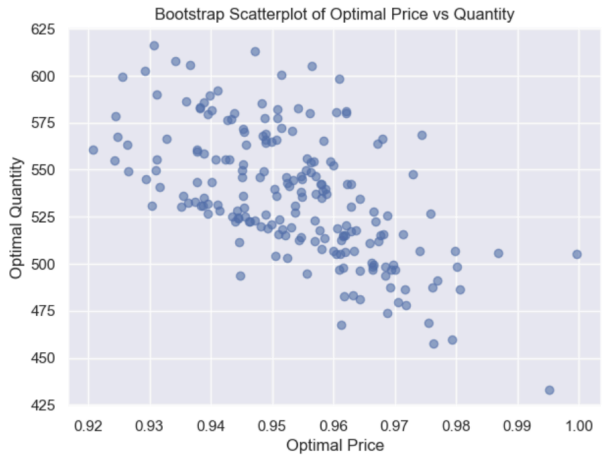
Histogram of Optimal Quantities.

The optimal quantities cluster around approximately 535 units, with very little dispersion, confirming the consistency of the production decision.

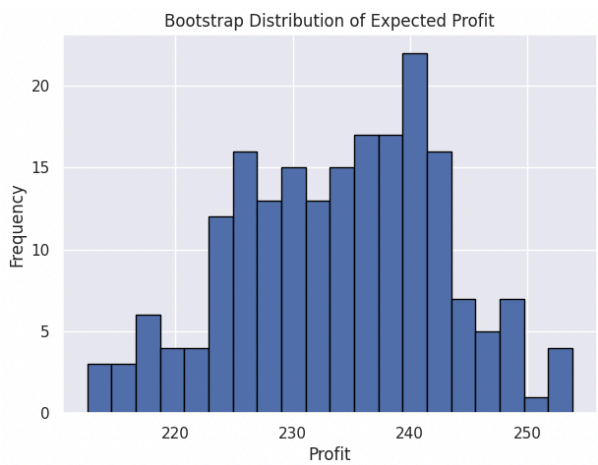


Joint Scatterplot with Marginal Histograms.

The scatterplot of optimal price versus optimal quantity forms a tight cluster, similar to Figure 8.3 in the report. This shows that the extended model repeatedly selects solutions in the same region of the price–quantity space, even as the underlying data changes.



Histogram of Expected Profits.
The profit distribution centers around $234.5, and the majority of values lie to the right of the standard newsvendor model's profit benchmark of $231.48. This mirrors the pattern illustrated in Figure 8.2.



7.3 Interpretation

These visualizations show that the extended model generates decisions that are both stable and consistently superior to the standard newsvendor model. The tight clustering of price and quantity indicates that the model captures a reliable economic relationship between price and demand. The profit histograms further demonstrate that the joint price–quantity optimization almost always improves expected performance, supporting the extended model's practical value.

Task 8. Evaluation of the Standard Newsvendor Model and the Extended Model

8.1 Is the Current Model as Good as the Extended Model?

When we apply the standard newsvendor model to the dataset at a fixed price of one dollar, it recommends printing about 472 units and produces an expected profit of $231.48. That decision is expected since the standard model only focuses on historical demand values and does not take into account how price affects customer behavior. Once we allow demand to respond to price, the decision shifts with the extended model recommending a price close to 0.95 and a print quantity of roughly 535 units, which increases expected profit to $234.42.

Figure 8.1: Demand vs. Price with Regression Curve and Confidence Band

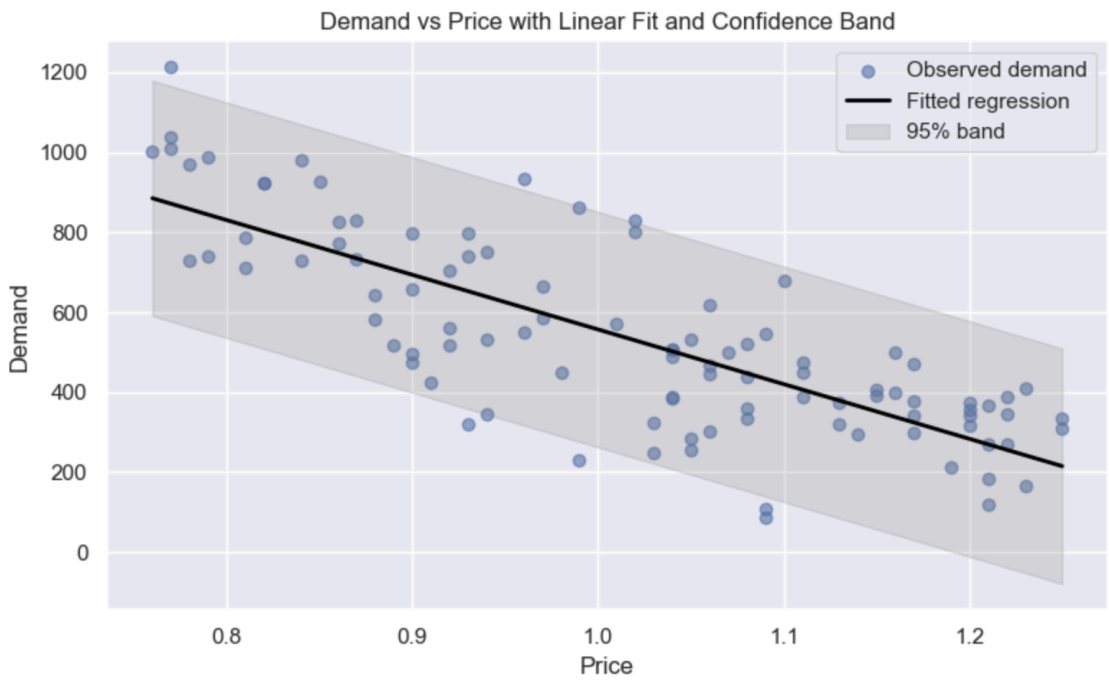


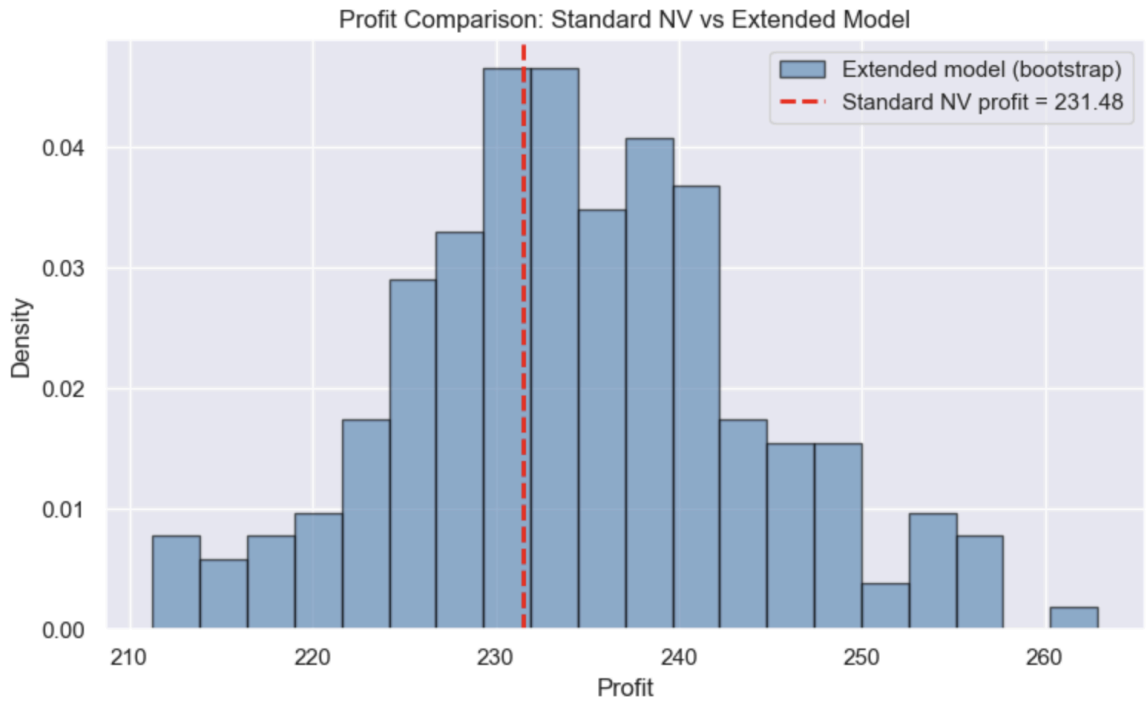
Figure 8.1 shows that demand decreases in a predictable way as price rises and the confidence band is narrow enough to show that this is not just due to random noise. This pattern matters because price is not just a number we randomly pick; it influences how many copies we expect to sell, which then affects the cost of running out of stock or ending up with excess inventory. The extended model uses this price–demand relationship directly, while the standard model behaves as if customers do not react to price at all. This difference leads to different decisions, and the extended model's decisions line up more closely with what the data suggest customers actually do. Therefore, we recommend using the extended model as it provides a better representation of the environment we are operating in, and it outperforms the standard model when applied to the same data.

8.2 Revenue Impact of Changing Models

We re-estimated the regression and resolved the optimization problem across 200 bootstrap samples in order to evaluate the stability of our results. Across these samples, the recommended price stayed very close to 0.95 and the print quantity remained tightly

concentrated around 535 units. The average profit across all samples was approximately $234.66, which is almost identical to the result we got from the original dataset.

Figure 8.2: Profit Distribution of Extended Model with Standard Model Benchmark



We can see the difference in revenue potential more clearly in Figure 8.2 as the distribution of profits from the extended model lies to the right of the standard newsvendor model's expected profit, which is shown as a vertical reference line. It is evident that the extended model does not just outperform the standard model in one particular scenario; it does so across a wide range of resampled datasets. This consistency shows that the extended model is capturing a real signal in the data rather than reacting to a specific sample. This pattern also suggests that the improvement is coming from the way the extended model uses structure in the data rather than from simply adjusting the print quantity upward. The model recognizes that price influences demand in a stable and predictable way, and that relationship creates a more favorable balance between expected sales and the costs of over- and under-production.

Figure 8.3: Joint Density of Bootstrapped Optimal Price and Quantity

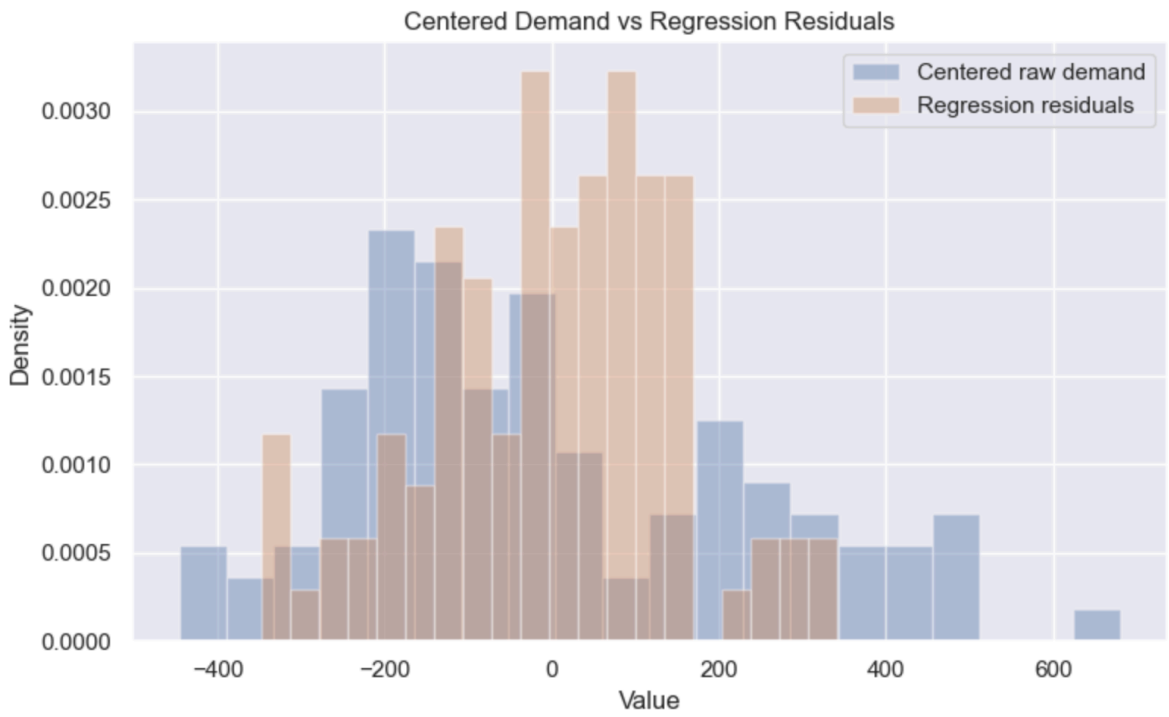


Figure 8.3 illustrates the joint density of the optimal price and optimal quantity across the bootstrap runs. The two decisions move together in a concentrated cluster, which shows that the extended model consistently identifies the same region of the price–quantity space even when the underlying data changes. This tells us that the model is not simply reacting to random fluctuations in individual demand observations; it is uncovering a stable economic relationship of how price affects demand and how the cost tradeoffs behave. This stability has direct revenue implications because the extended model returns similar decisions across many samples, which reduces the likelihood of overreacting to short-term noise. As a result, sales volumes become more predictable and the number of costly mismatches between print quantity and actual demand declines. Since rush printing and disposal costs are asymmetric, even small deviations from the true decision can create meaningful losses, but the extended model's tight clustering minimizes those deviations. The overall benefit is that we would see more periods end with quantities that match demand closely enough to avoid both excess inventory and costly stockouts.

8.3 Advantages and Disadvantages of Both Models

The standard newsvendor model has the benefit of being simple to run and easy to interpret, as it only requires past demand observations and a few basic cost parameters, making it practical in situations where decisions must be made quickly or when data is limited. The drawback of this model is that its simplicity forces the model to ignore patterns that are clearly present in the dataset. When the model treats demand as independent of price, it overlooks the clear relationship in Figure 8.1, and this prevents price from playing any role in the decision. That restriction pushes the solution toward quantities that mirror historical averages rather than the

changes in demand that occur when price shifts. The model also fails to account for the different operational consequences of underproduction and overproduction. For example, rush printing and disposal costs behave differently; however, the standard model treats them as equal and collapses those distinctions into a single decision rule. As a result, its recommendations may appear intuitive, but they actually don't fully capture the economic tradeoffs.

The extended model addresses these constraints by incorporating the structure that we observe directly in the data and uses a regression-based demand function, which allows the decision to account for the way price influences expected sales. Once price becomes part of the model, a substantial portion of the variability in demand becomes predictable, as shown by the tighter distribution of regression residuals in Figure 8.4. This improves the model's ability to align the print quantity with expected demand and reduces the likelihood of facing excess inventory or a stockout. The extended model also evaluates price and quantity at the same time, which allows it to reflect the true tradeoffs among revenue, rush-printing penalties, and disposal costs rather than relying on simplified assumptions. It is important to note that the extended model takes more resources and effort to run, and it depends on available data to estimate the price-demand relationship. This model also benefits from periodic updates since it is common for the underlying demand pattern to evolve. These requirements allow the extended model to produce decisions that are better aligned with the data and, as shown in the bootstrap analysis, lead to higher and more stable expected profits. The extended model adds substantial value in the long term since printing and pricing decisions are made frequently, and the model can adjust to new data as those decisions accumulate.

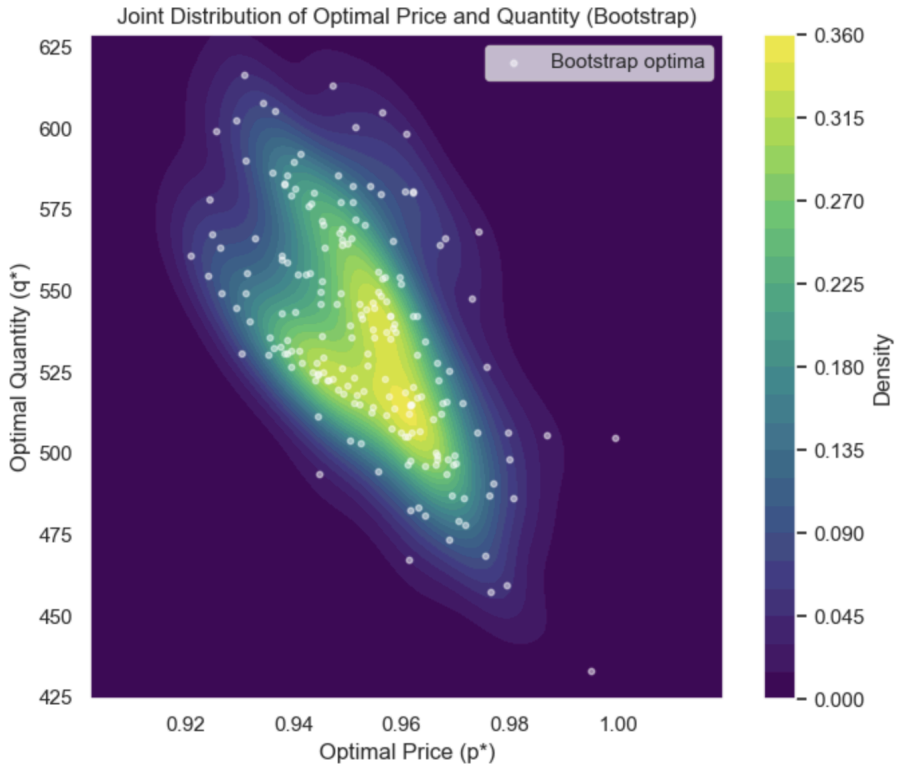



Figure 8.4: Distribution of Centered Demand and Regression Residuals